

Vignesh Chowdary Pamulapati

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Education

Northern Arizona University <i>Master of Science - Computer Science; GPA: 3.88 / 4.0</i>	Arizona, United States Aug 2024 – Dec 2025
Keshav Memorial Institute of Technology <i>Bachelor of Technology - Computer Science in AIML Specialization; GPA: 8.14 / 10</i>	Hyderabad, India June 2020 - April 2024

Experience

Northern Arizona University <i>Teaching Assistant</i>	Arizona, United States Jan 2025 – May 2025
- Mentored 30+ students in Data Structures and Python, strengthening understanding of core algorithms, programming concepts, and problem-solving techniques, contributing to a 40% improvement in course performance. - Developed a reusable algorithm template library, reducing debugging time by 30% across student submissions. - Designed and integrated dynamic UI components (dashboards, interactive coding modules, automated feedback systems) improving student engagement by 25%. - Partnered with faculty members to refine course content and strengthen the alignment between coursework and practical software engineering practices.	

Nowfloats Technologies <i>Software Engineering Intern</i>	Hyderabad, India July 2023 - Dec 2023
- Contributed to developing a file-system-based mini-database engine in C by implementing page management, record storage, B+ Tree indexing, and a lightweight query parser, improving overall query throughput by 35%. - Assisted in developing REST APIs and backend modules for bizHQ/Boost360 using Node.js and Express, implementing CRUD endpoints and contributing to user workflow features used across web and Android clients. - Contributed to developing modular, reusable UI components in React/Bootstrap for analytics dashboards and business insights, improving usability and reducing UI load time. - Worked with the backend team to optimize API performance by profiling latency issues and updating inefficient queries, improving responsiveness for end users. - Wrote unit tests, participated in code reviews, and contributed to Agile sprints, ensuring stable release cycles and high code quality.	

Projects

AI Personal Knowledge Base – RAG-Driven Document Query Platform <i>Tech - React, TypeScript, Node.js, Express, MongoDB, Google Gemini AI</i>	
- Built an AI-powered personal knowledge base enabling document upload and natural-language querying through a retrieval-augmented generation (RAG) pipeline using Gemini AI and vector embeddings. - Engineered backend workflows for document parsing, text chunking, embedding generation, and vector similarity search with Node.js, Express, and MongoDB for scalable, low-latency semantic retrieval. - Developed an interactive chat interface in React/TypeScript supporting context-grounded AI responses, citation of source text, and persistent conversation history.	
LinguaLink – AI-Powered Multilingual Communication Platform <i>Tech - React, TypeScript, Tailwind CSS, Node.js, Framer Motion, Rest API's, Socket.IO, MongoDB</i>	
- Engineered a real-time multilingual communication platform using Socket.IO and Google Gemini AI, enabling sub-second bilingual chat and translation across 15+ languages. - Designed a scalable backend architecture (Node.js/Express + MongoDB) with translation caching and optimized schemas, reducing redundant AI requests by ~40% and improving overall system responsiveness. - Implemented secure collaborative workspaces with role-based access control, group channels, and direct messaging using REST APIs and event-driven WebSocket communication. - Developed a modular, responsive frontend with React/TypeScript, Tailwind, and Framer Motion, leveraging Context API for unified state management across authentication, workspaces, and UI themes.	

Technical Skills

Programming: Python, C++, JavaScript, TypeScript, Java, C, SQL, Data Structures and Algorithms
Frontend: React.js, HTML5, CSS3, Tailwind
Backend: Node.js, Express.js, Flask, REST APIs
Databases: MongoDB, MYSQL, PostgreSQL
Cloud & DevOps: AWS (EC2, Lambda), Microsoft Azure, Git, Github
Practices: Agile, Code Review, Software Design Principles